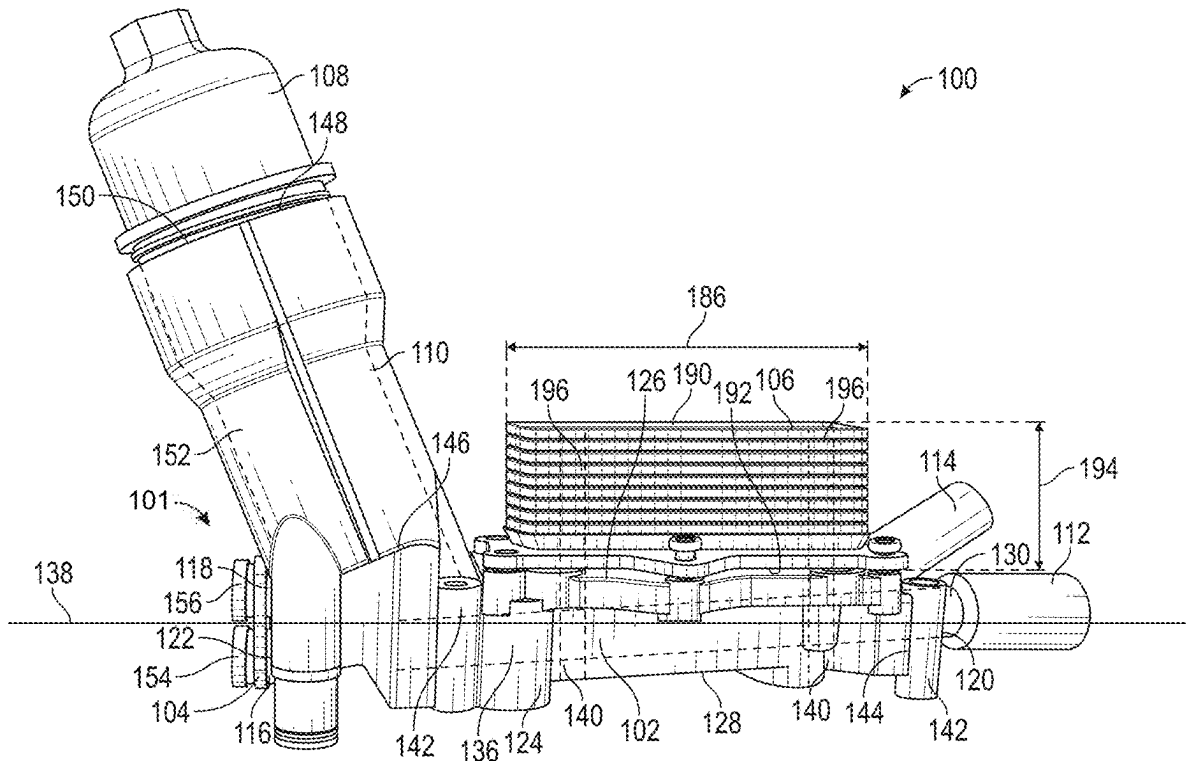




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(19) **United States**(12) **Patent Application Publication****Le et al.**(10) **Pub. No.: US 2025/0277462 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **OIL FILTER HOUSING AND ASSEMBLY****Publication Classification**(71) Applicant: **Skyward Automotive Products LLC**,
Tolleson, AZ (US)(51) **Int. Cl.**
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Gilbert, AZ (US)(52) **U.S. Cl.**
CPC **F01M 11/03** (2013.01); **F01M 2011/033**
(2013.01)(73) Assignee: **Skyward Automotive Products LLC**,
Tolleson, AZ (US)(57) **ABSTRACT**(21) Appl. No.: **18/820,661**(22) Filed: **Aug. 30, 2024**

An oil filter assembly comprising an oil filter housing having an oil filter housing first portion having an oil filter housing first portion first end, an opposing oil filter housing first portion second end, an oil filter housing first portion first surface and an opposing oil filter housing first portion second surface; an oil filter housing second portion having an oil filter housing second portion first surface, an oil filter housing second portion second surface opposing the oil filter housing second portion first surface, and a perimeter wall extending fully around the oil filter housing second portion and orthogonally away from the oil filter housing second portion first surface; an oil filter; and an oil cooler.

Related U.S. Application Data(63) Continuation of application No. 18/591,849, filed on
Feb. 29, 2024, now Pat. No. 12,078,090.

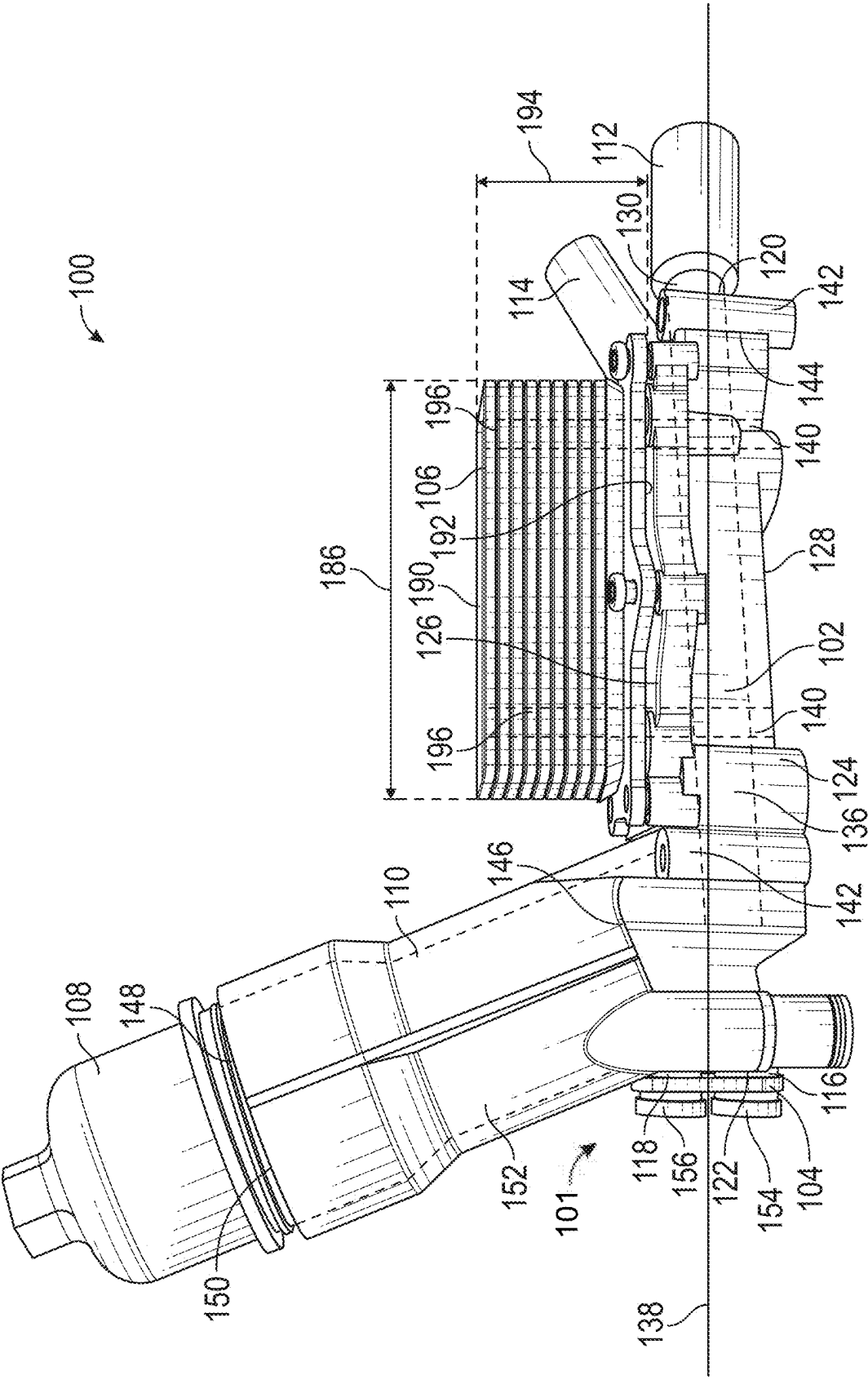
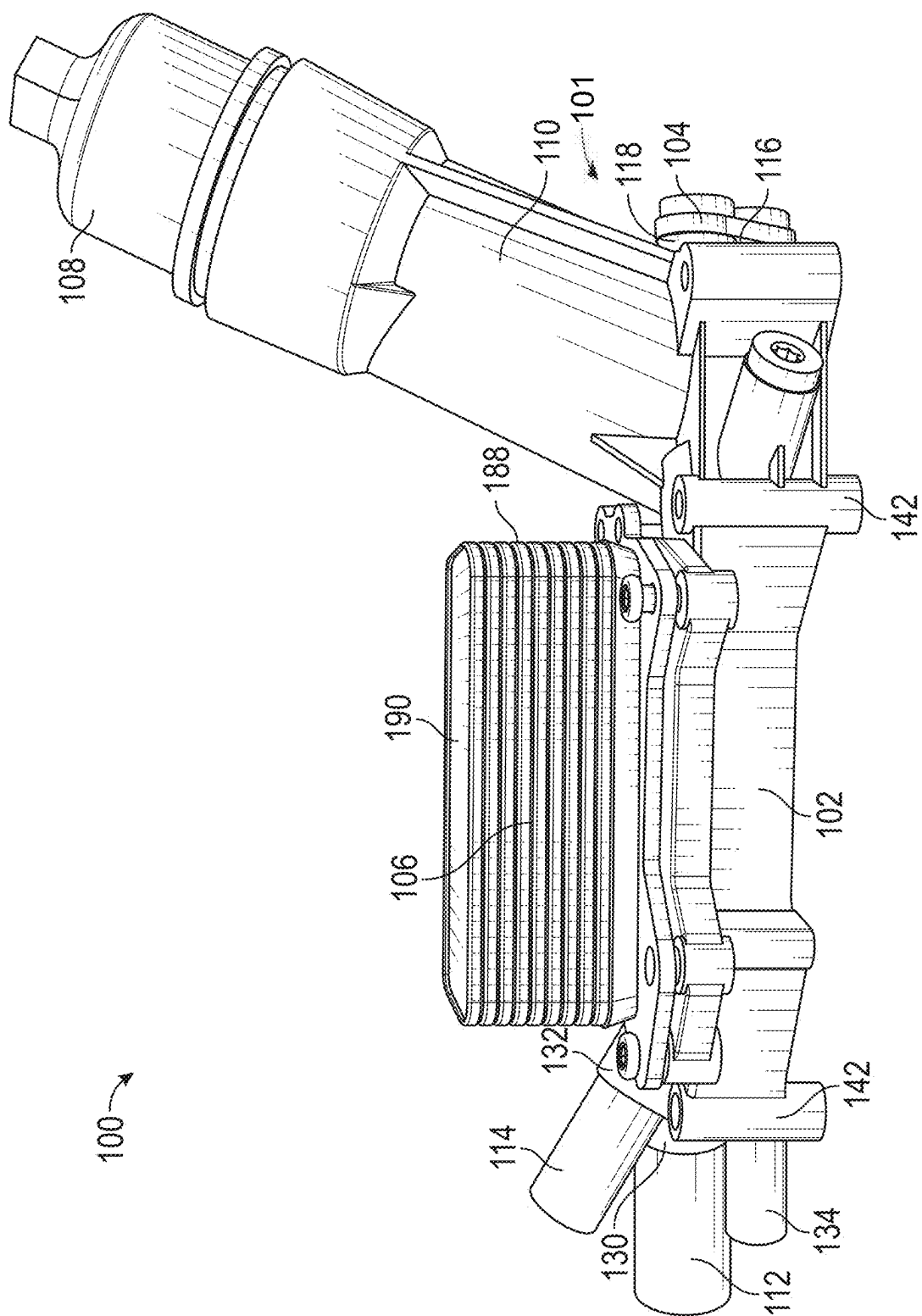


FIG. 1



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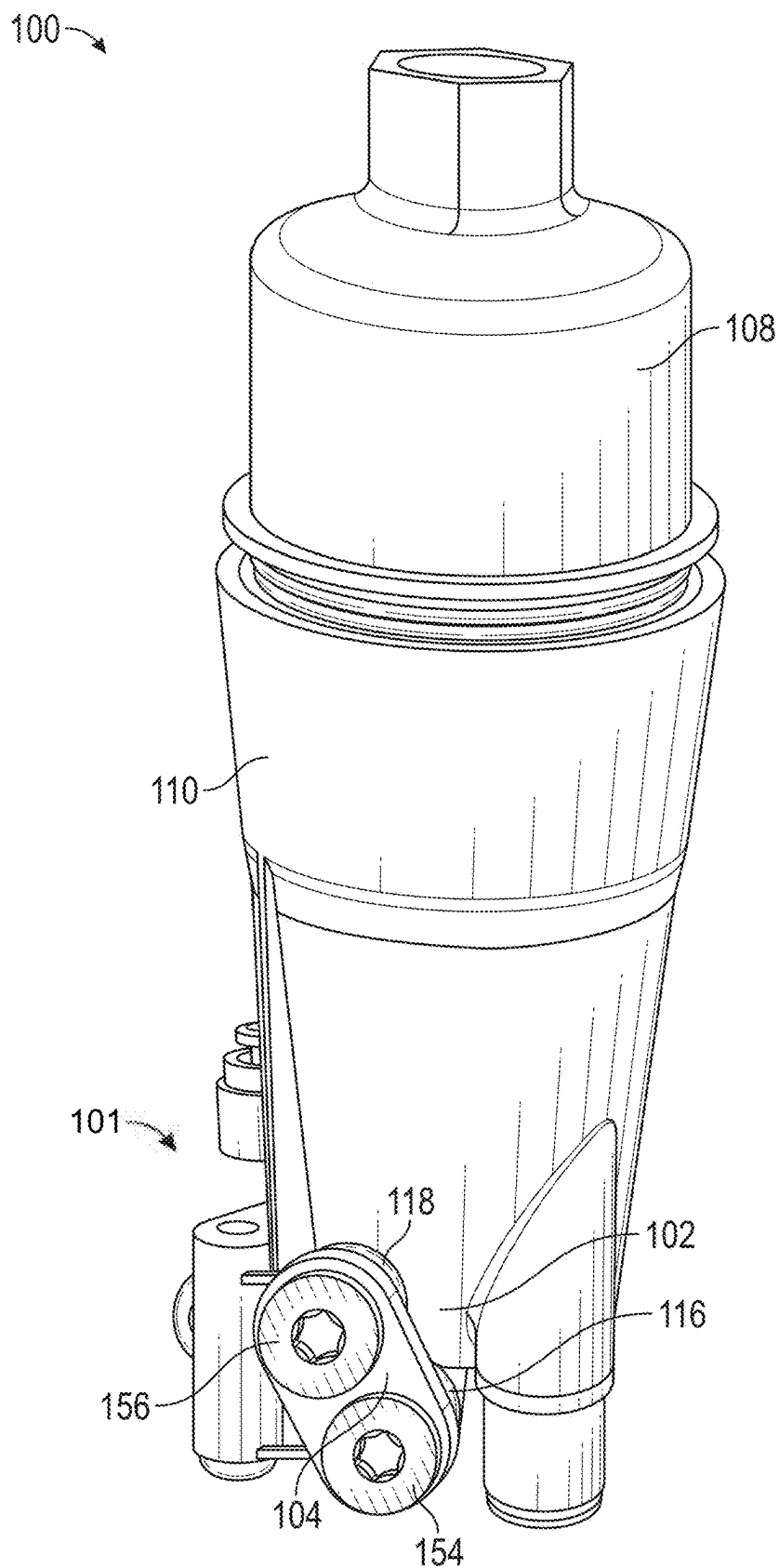


FIG. 3

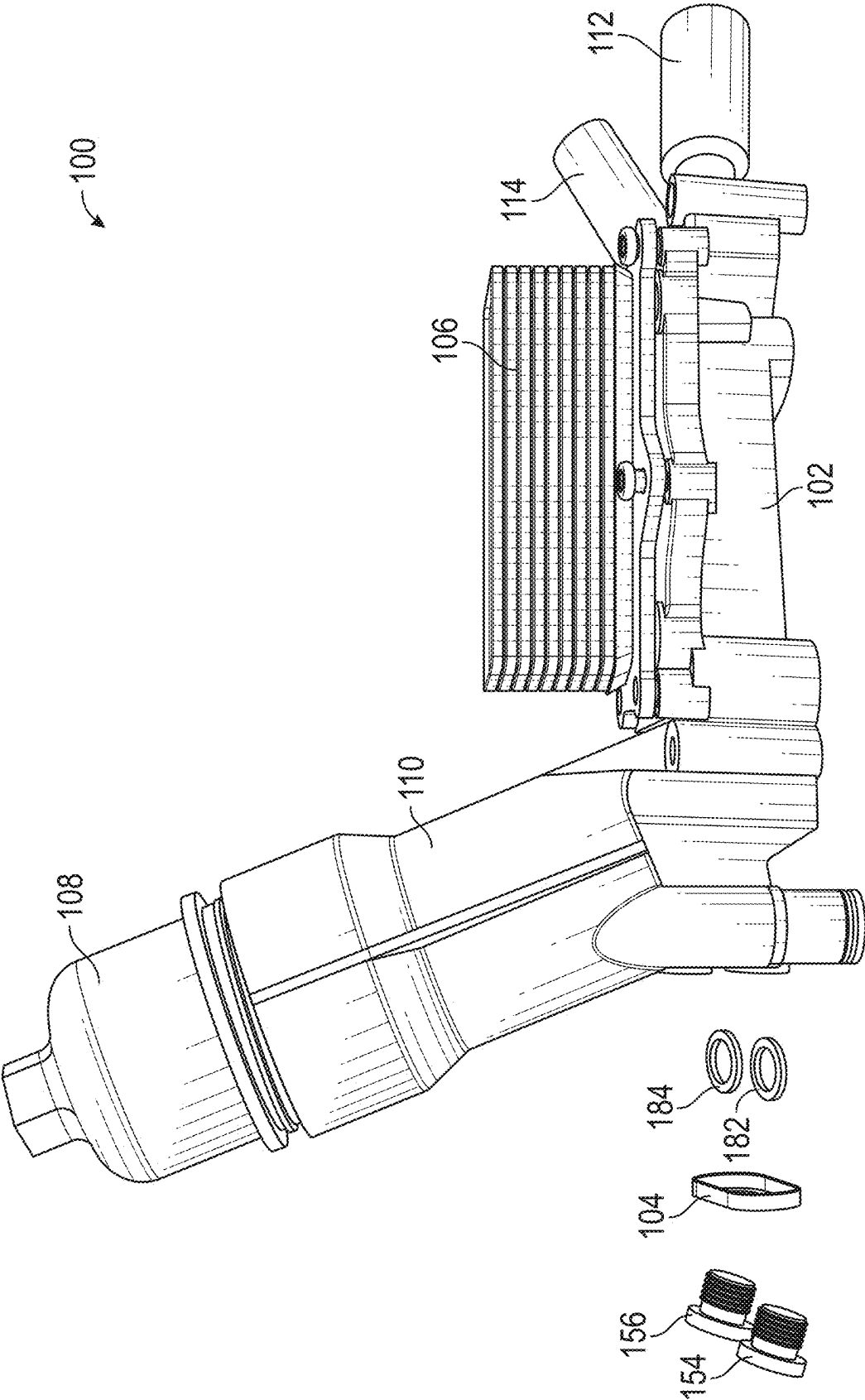


FIG. 4

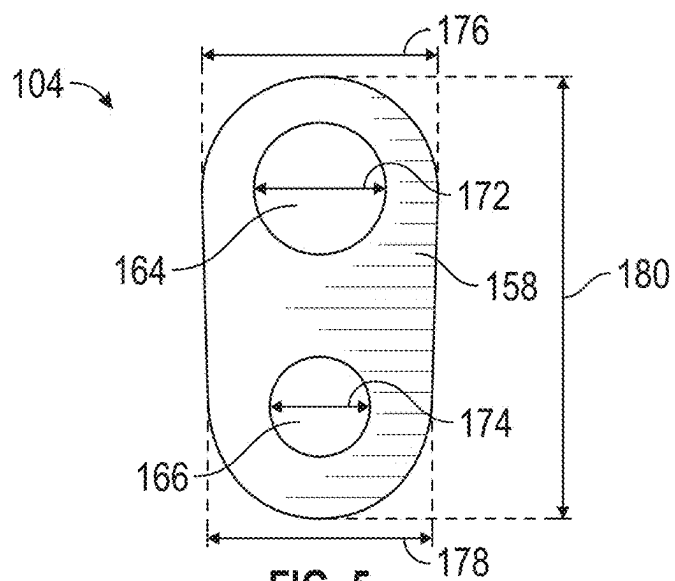


FIG. 5

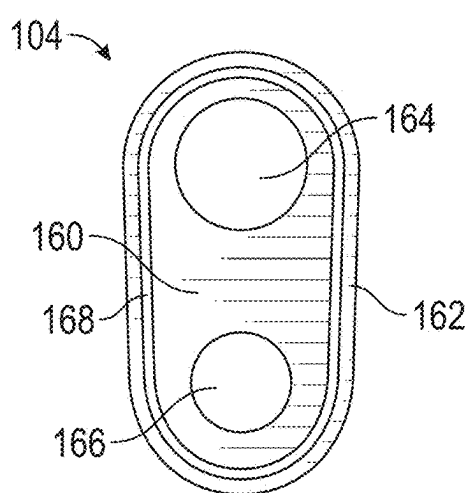


FIG. 6

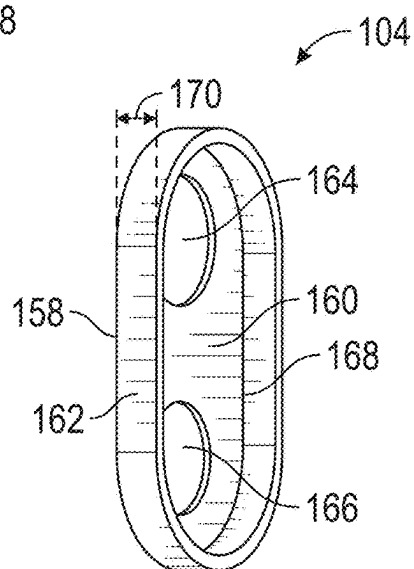


FIG. 7

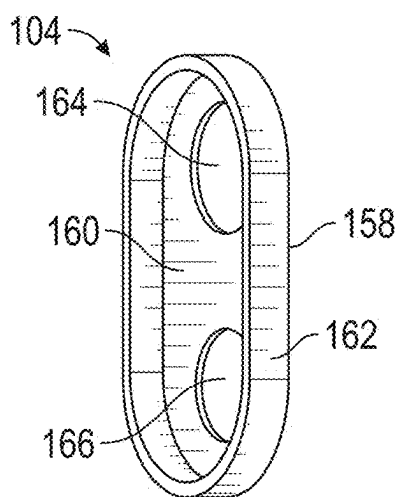


FIG. 8

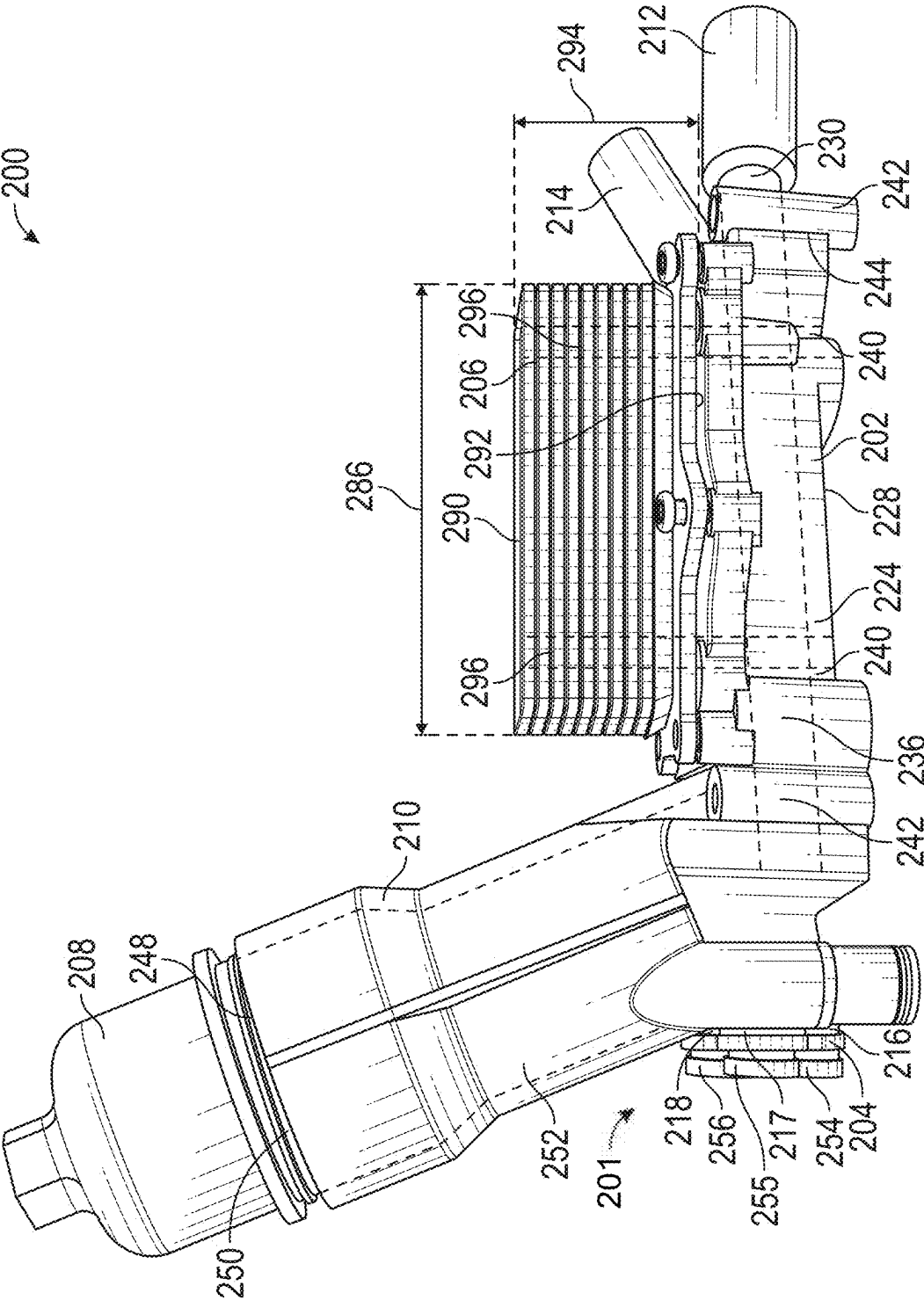


FIG. 9

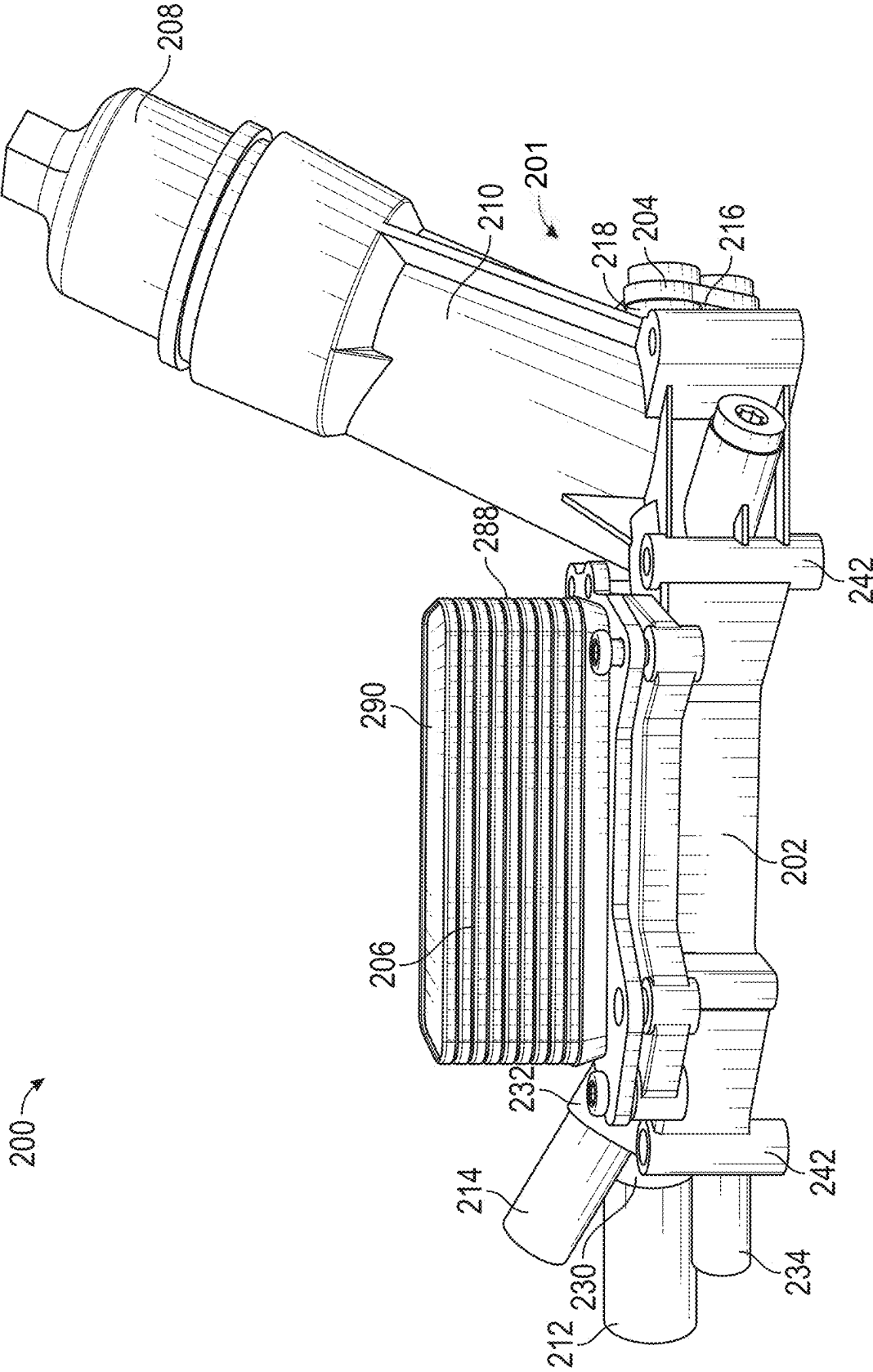


FIG. 10

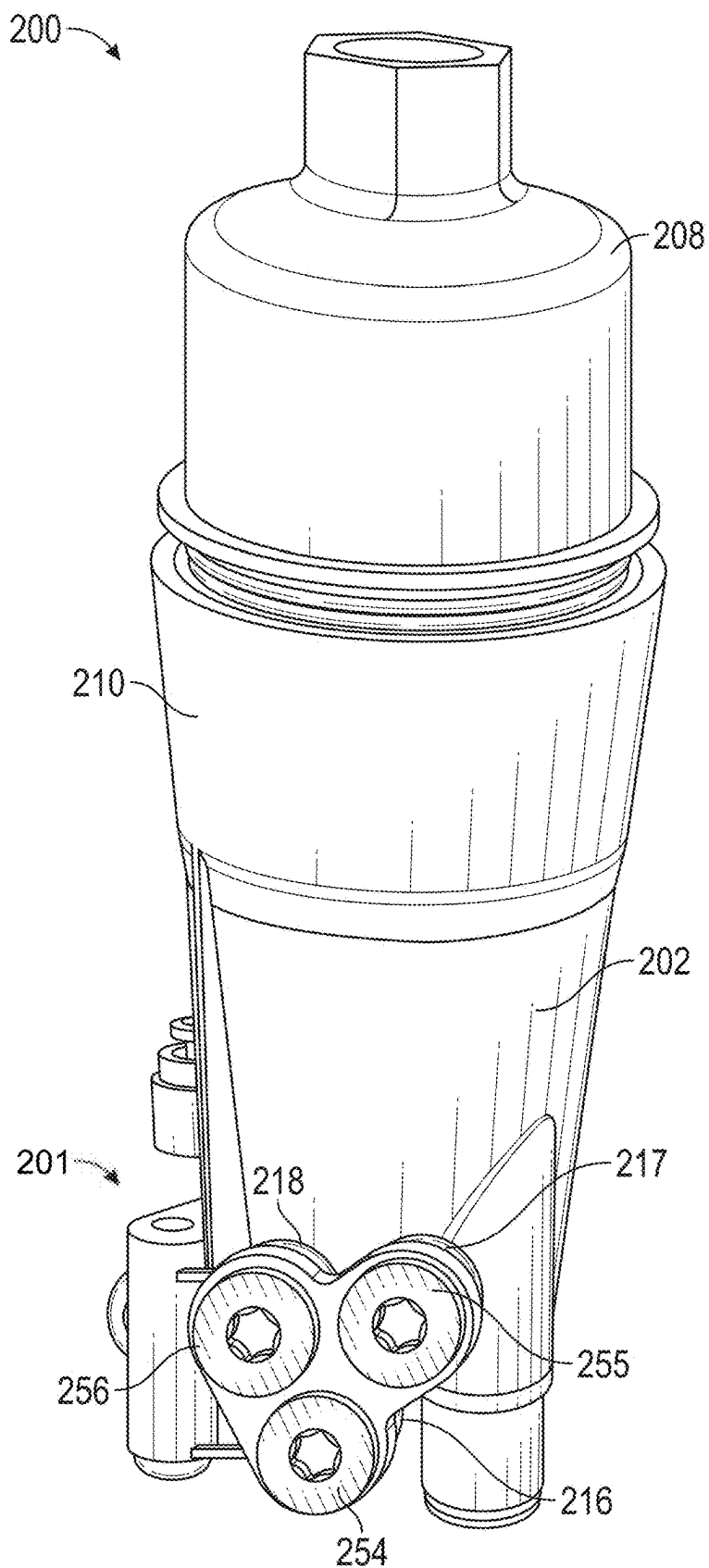


FIG. 11

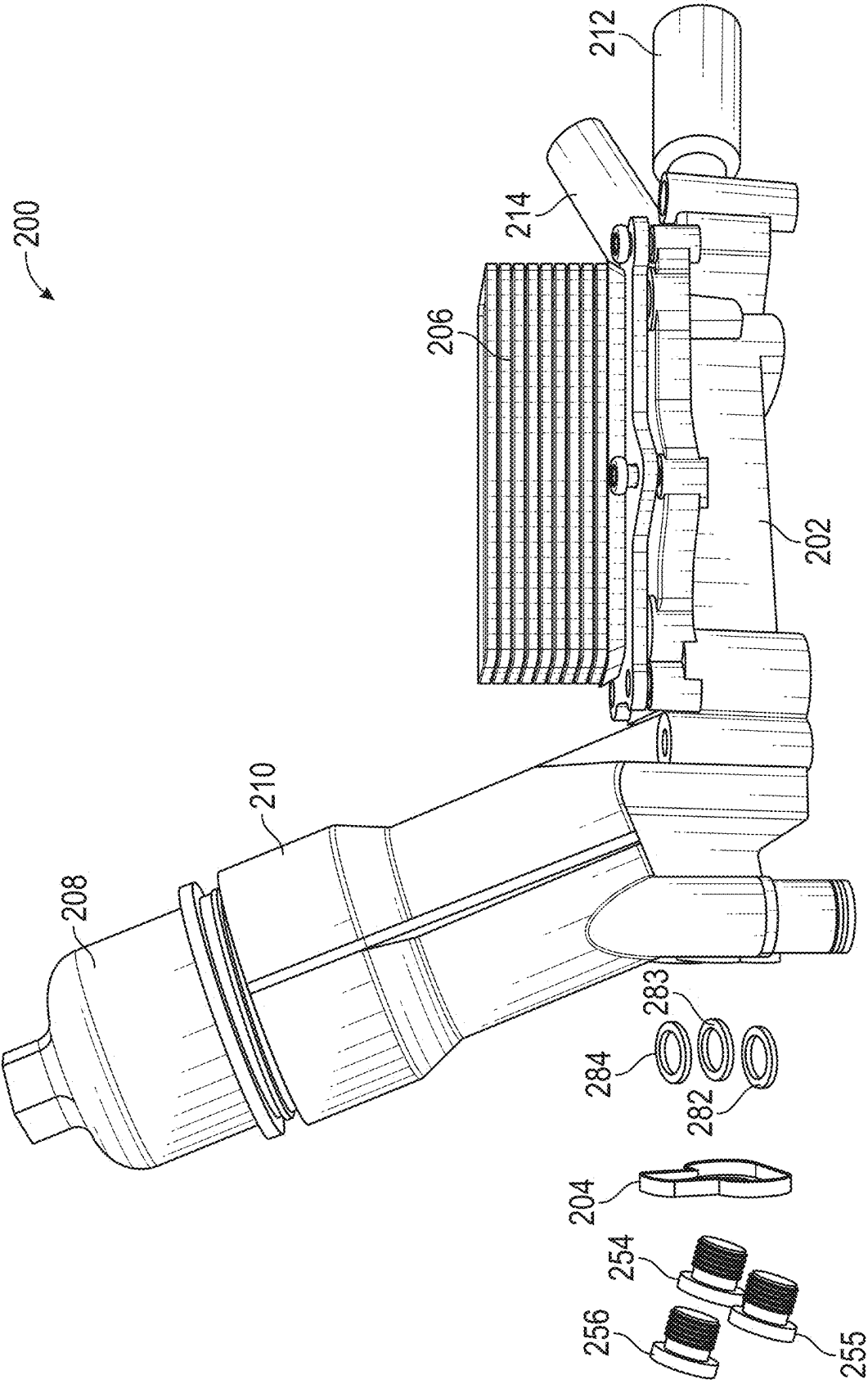


FIG. 12

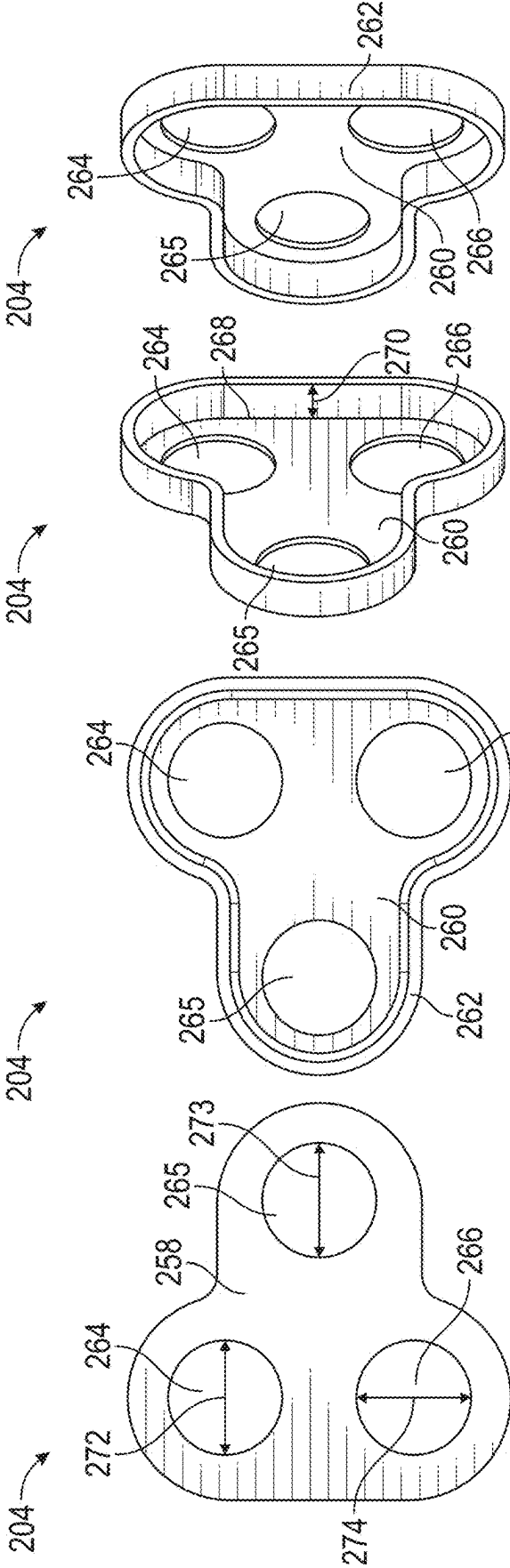
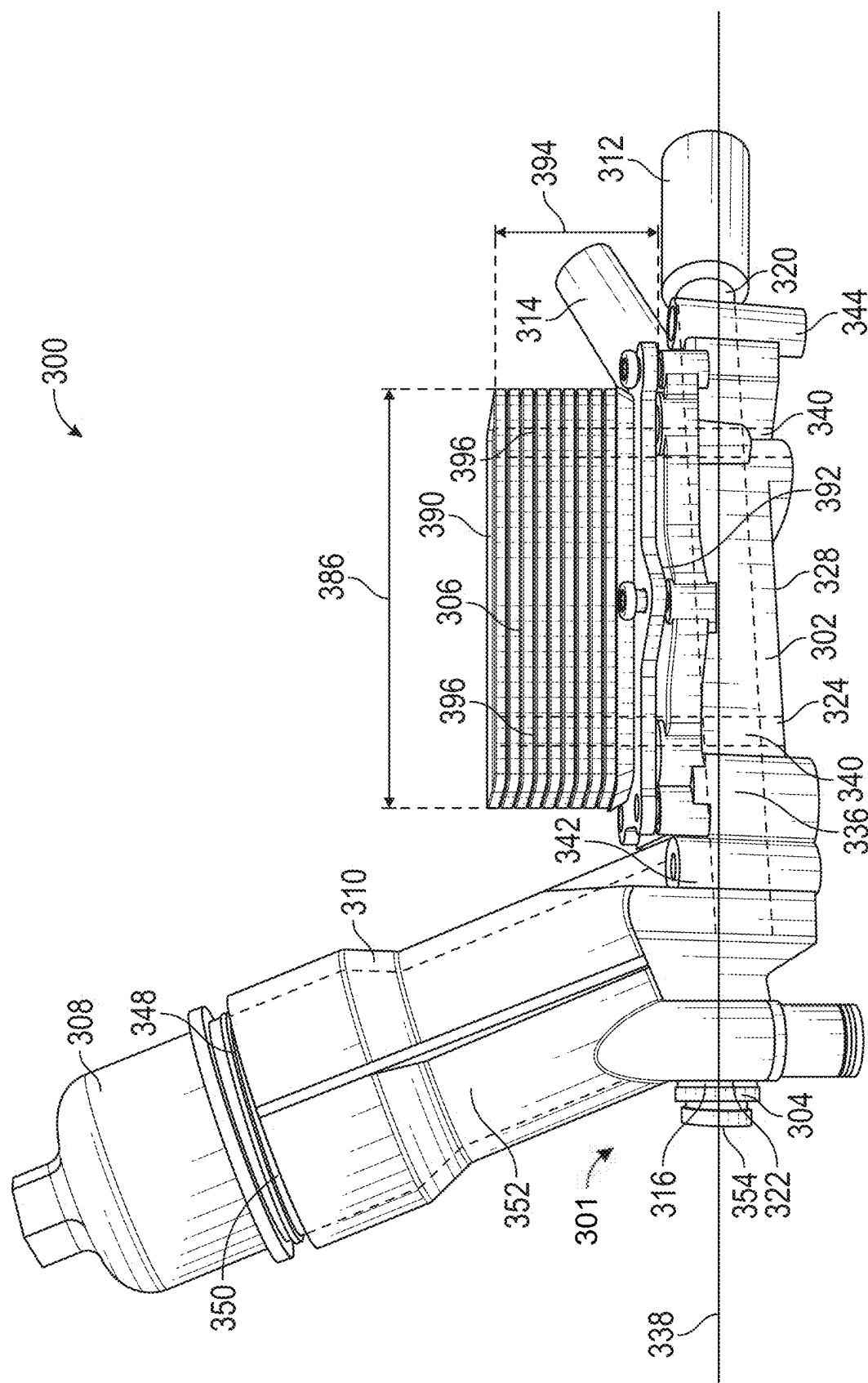


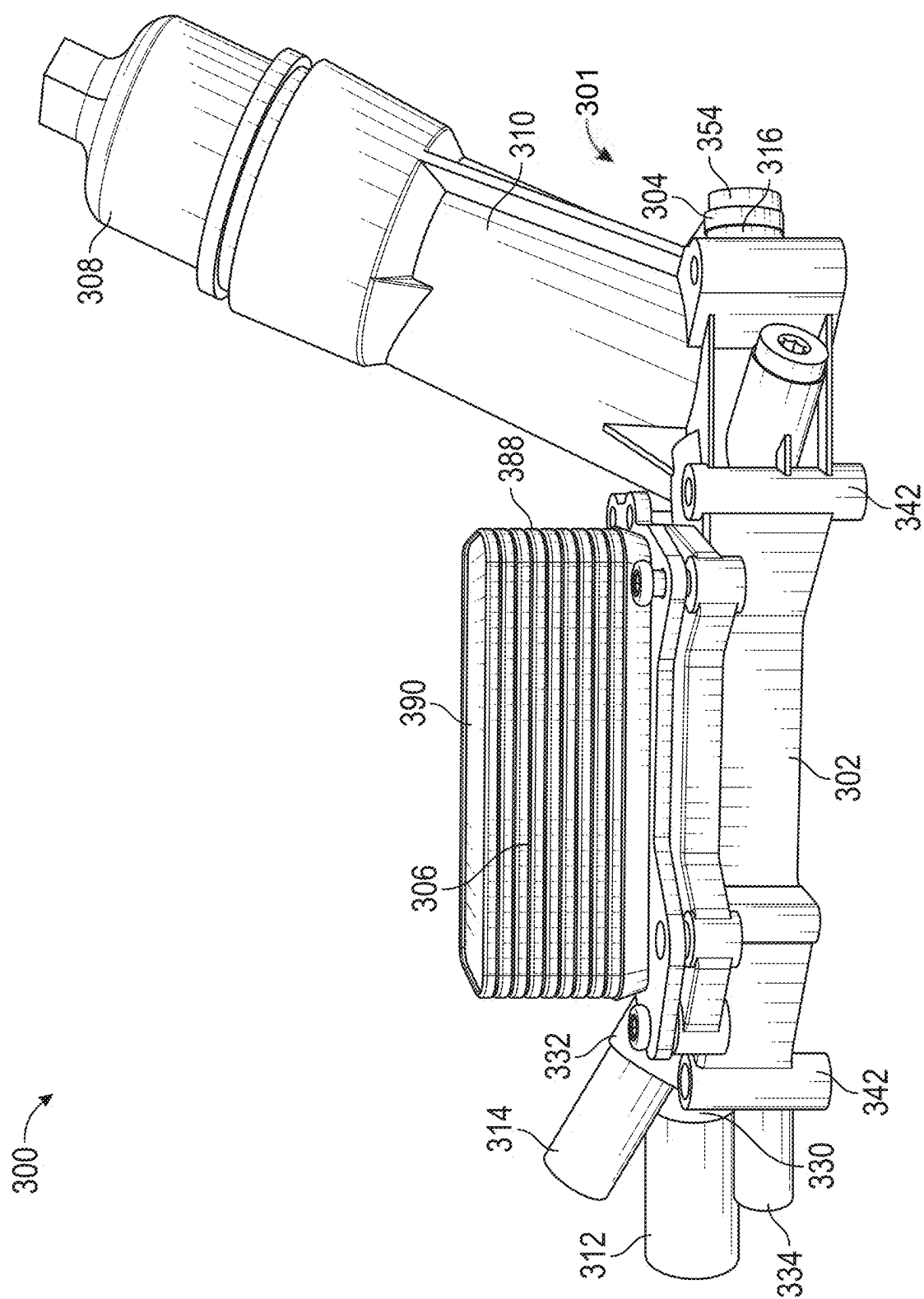
FIG. 13

FIG. 14

FIG. 15

FIG. 16




$$\frac{\infty}{\sqrt{\infty}} \times \frac{\infty}{\infty} = \frac{\infty}{\infty}$$

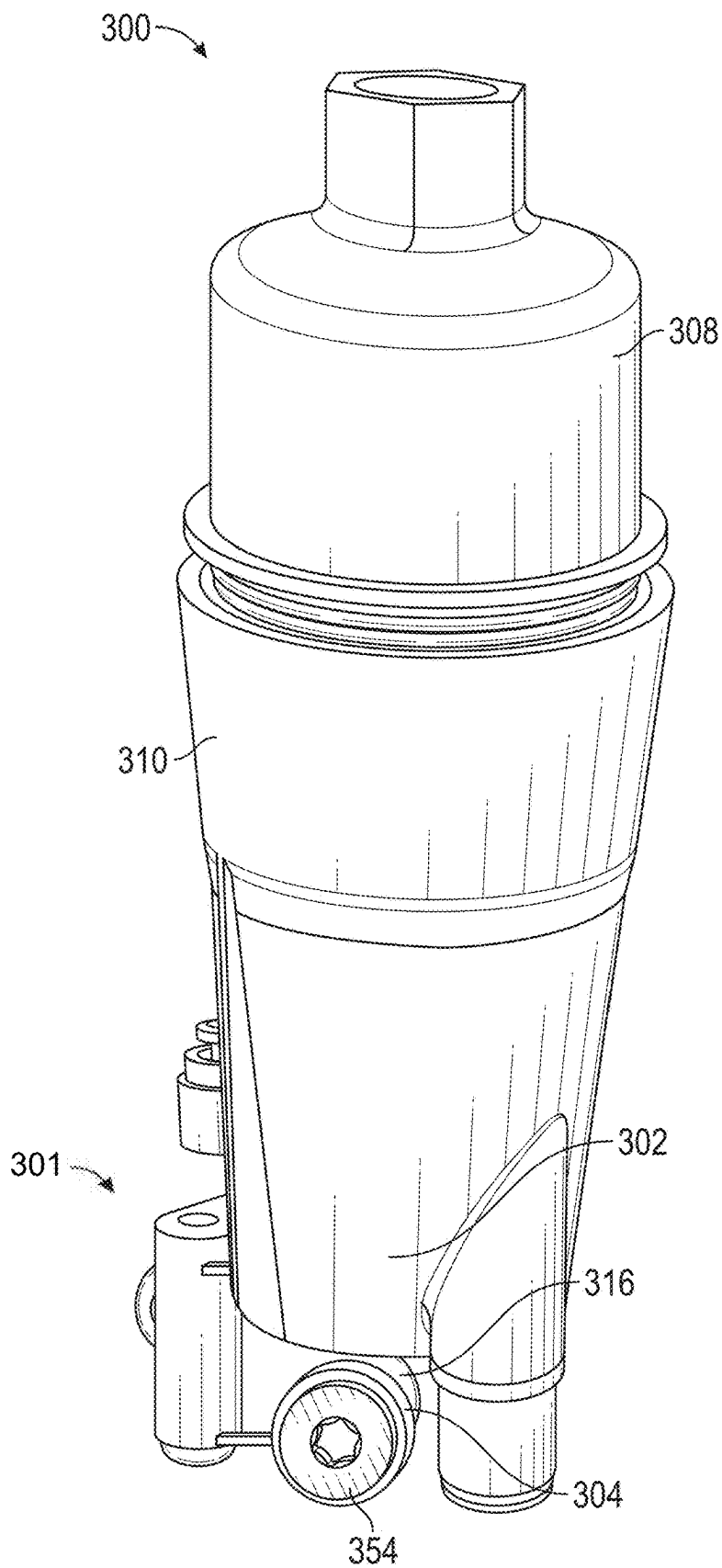


FIG. 19

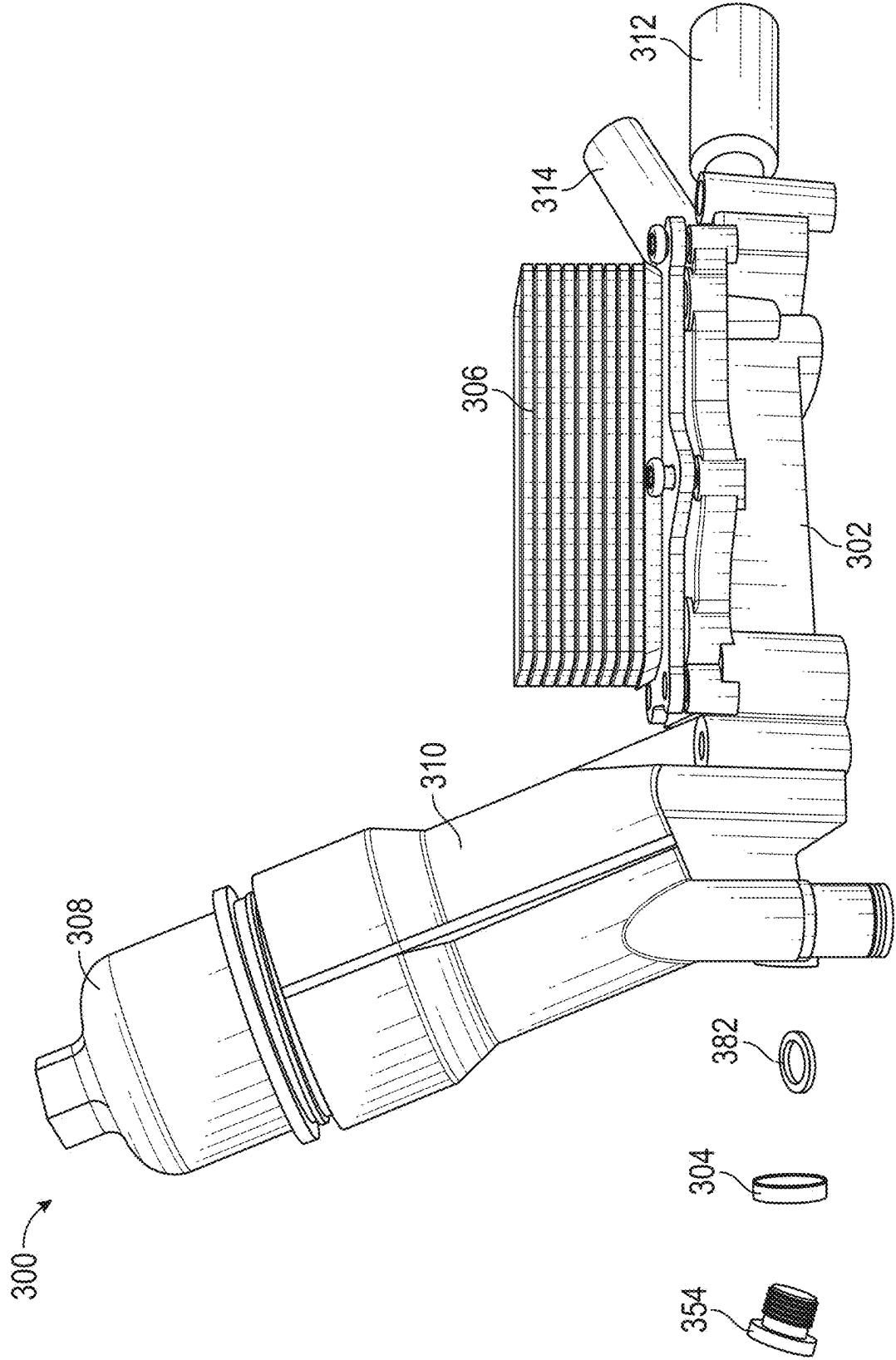


FIG. 20

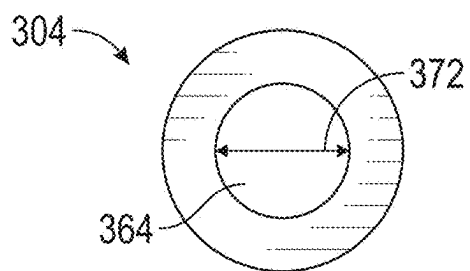


FIG. 21

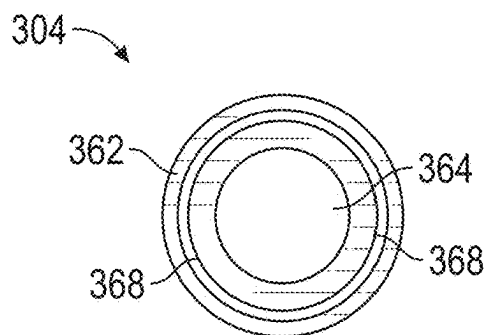


FIG. 22

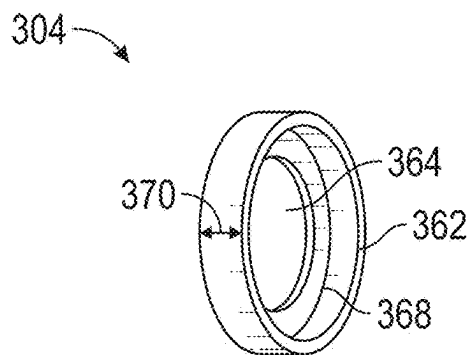


FIG. 23

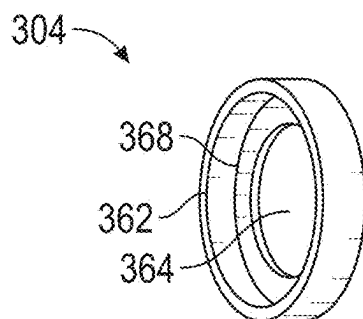


FIG. 24

OIL FILTER HOUSING AND ASSEMBLY

FIELD

[0001] The disclosure relates to internal combustion engines. More particularly, this disclosure relates to oil filter housings and assemblies that include an oil filter housing. The oil filter housings and assemblies are useful in filtering and cooling lubrication oil used in internal combustion engines.

BACKGROUND

[0002] Oil filter housings are critical components of some internal combustion engines. Oil filter assemblies typically include an oil filter and an oil cooler secured to a housing component. Multiple conventional oil filter assemblies include a housing component that is a single, unitary component. In other words, the housing component of several conventional oil filter assemblies has a unitary construction comprising only a single part, which can be formed of plastic or cast as a metal part. While these unitary housing components may simplify manufacturing processes, they, ironically, provide multiple distinct points of failure by including multiple distinct flow ports that lack a common structure away and distinct from the unitary housing component.

[0003] A need remains, therefore, for improved oil filter housings and oil filter assemblies that include an improved oil filter housing.

BRIEF SUMMARY OF SELECTED EXAMPLES

[0004] Various example oil filter housings and oil filter assemblies are described.

[0005] An example oil filter assembly comprises an oil filter housing having an oil filter housing first portion having an oil filter housing first portion first end and an opposing oil filter housing first portion second end, an oil filter housing first portion surface and an opposing oil filter housing first portion second surface, the first portion first end defining an oil filter tube, the second end defining first and second flow ports, the first surface adapted for a secured connection with an internal combustion engine; an oil filter housing second portion having an oil filter housing second portion first surface, an oil filter housing second portion second surface opposing the oil filter housing second portion first surface, a perimeter wall extending fully around the oil filter housing second portion and orthogonally away from the oil filter housing second portion first surface, the oil filter housing second portion defining first and second holes extending through the oil filter housing second portion from the oil filter housing second portion first surface to the oil filter housing second portion second surface, the oil filter housing second portion disposed on the oil filter housing first portion such that the first flow port extends through the first hole, the second flow port extends through the second hole, and the perimeter wall extends toward the oil filter housing first portion; an oil filter adapted to be disposed within the oil filter tube; and an oil cooler adapted to be secured to the oil filter housing first portion first surface.

[0006] Another example oil filter assembly comprises an oil filter housing having an oil filter housing first portion having an oil filter housing first portion first end, an opposing oil filter housing first portion second end, an oil filter housing first portion first surface, and an opposing oil filter

housing first portion second surface, the oil filter housing first portion first end defining an oil filter tube, the oil filter housing first portion second end defining a flow port, the oil filter housing first portion first surface adapted for a secured connection with an internal combustion engine; an oil filter housing second portion having an oil filter housing second portion first surface, an oil filter housing second portion second surface opposing the oil filter housing second portion first surface, a perimeter wall extending fully around the oil filter housing second portion and orthogonally away from the oil filter housing second portion first surface, the oil filter housing second portion defining a hole extending through the oil filter housing second portion from the oil filter housing second portion first surface to the oil filter housing second portion second surface, the oil filter housing second portion disposed on the oil filter housing first portion such that the flow port extends through the hole, and the perimeter wall extends toward the oil filter housing first portion; an oil cooler having an oil cooler first surface and an oil cooler second surface removably attached to the oil filter housing first portion first surface; and an oil filter adapted to be disposed within the oil filter tube.

[0007] Another example oil filter assembly comprises an oil filter housing having an oil filter housing first portion having an oil filter housing first portion first end, an opposing oil filter housing first portion second end, an oil filter housing first portion first surface and an opposing oil filter housing first portion second surface, the oil filter housing first portion first end defining an oil filter tube, the oil filter housing first portion second end defining a flow port, the first surface adapted for a secured connection with an internal combustion engine, the oil filter housing first portion having an oil filter housing first portion vertical channel extending from the oil filter housing first portion first surface to the oil filter housing first portion second surface; an oil filter housing second portion having an oil filter housing second portion first surface, an oil filter housing second portion second surface opposing the oil filter housing second portion first surface, a perimeter wall extending fully around the oil filter housing second portion and orthogonally away from the oil filter housing second portion first surface, the oil filter housing second portion defining a hole extending through the oil filter housing second portion from the oil filter housing second portion first surface to the oil filter housing second portion second surface, the oil filter housing second portion disposed on the oil filter housing first portion such that the flow port extends through the hole, and the perimeter wall extends toward the oil filter housing first portion; an oil cooler having an oil cooler first surface and an oil cooler second surface removably attached to the oil filter housing first portion first surface, the oil cooler having an oil cooler vertical channel extending from the oil cooler second surface in a direction toward the oil cooler first surface; and an oil filter adapted to be disposed within the oil filter tube.

[0008] Additional understanding of the inventive oil filter housings and oil filter assemblies can be obtained by reviewing the detailed description of selected examples, below, and the referenced drawings.

DESCRIPTION OF FIGURES

[0009] FIG. 1 is a first side view of a first example oil filter assembly.

[0010] FIG. 2 is a second side view of the first example oil filter assembly.

[0011] FIG. 3 is a third side view of the first example oil filter assembly.

[0012] FIG. 4 is a partially exploded view of the first side of the first example oil filter assembly having an oil filter housing second portion removed from an oil filter housing first portion.

[0013] FIG. 5 is a top view of the oil filter housing second portion.

[0014] FIG. 6 is a bottom view of the oil filter housing second portion.

[0015] FIG. 7 is a first perspective view of the oil filter housing second portion.

[0016] FIG. 8 is a second perspective view of the oil filter housing second portion.

[0017] FIG. 9 is a first side view of a second example oil filter assembly.

[0018] FIG. 10 is a second side view of the second example oil filter assembly.

[0019] FIG. 11 is a third side view of the second example oil filter assembly.

[0020] FIG. 12 is a partially exploded view of the first side view of the second example oil filter assembly having an oil filter housing second portion removed from an oil filter housing first portion.

[0021] FIG. 13 is a top view of the oil filter housing second portion.

[0022] FIG. 14 is a bottom view of the oil filter housing second portion.

[0023] FIG. 15 is a first perspective view of the oil filter housing second portion.

[0024] FIG. 16 is a second perspective view of the oil filter housing second portion.

[0025] FIG. 17 is a first side view of a third example oil filter assembly.

[0026] FIG. 18 is a second side view of the third example oil filter assembly.

[0027] FIG. 19 is a third side view of the third example oil filter assembly.

[0028] FIG. 20 is a partially exploded view of the first side view of the third example oil filter assembly having an oil filter housing second portion removed from an oil filter housing first portion.

[0029] FIG. 21 is a top view of the oil filter housing second portion.

[0030] FIG. 22 is a bottom view of the oil filter housing second portion.

[0031] FIG. 23 is a first perspective view of the oil filter housing second portion.

[0032] FIG. 24 is a second perspective view of the oil filter housing second portion.

DETAILED DESCRIPTION OF SELECTED EXAMPLES

[0033] The following detailed description and the appended drawings describe and illustrate various example oil filter housings and example oil filter assemblies. The description and illustration of these examples enable one skilled in the art to make and use example oil filter housings and oil filter assemblies. They do not limit the scope of the claims in any manner.

[0034] Each of FIGS. 1, 2, 3, and 4 illustrates a first example oil filter assembly 100. The oil filter assembly 100 includes an oil filter housing 101, an oil filter (not illustrated) disposed in an oil filter tube 110 defined by the oil filter

housing 101, and an oil cooler 106. The oil filter housing 101 has an oil filter housing first portion 102 and an oil filter housing second portion 104. The oil filter assembly 100 further has a cap 108, a first electrical connector 112, a second electrical connector 114, a first flow port 116, and a second flow port 118.

[0035] As illustrated, the oil filter housing first portion 102 has an oil filter housing first portion first end 120, oil filter housing first portion second end 122, and an oil filter housing first portion intermediate portion 124 extending between the oil filter housing first portion first end 120 and the oil filter housing first portion second end 122. The oil filter housing first portion 102 further has an oil filter housing first portion first surface 126 and an oil filter housing first portion second surface 128. The oil filter housing first portion first end 120 comprises a first port 130 and a second port 132. The oil filter housing first portion first end 120 can have as many ports as desired including, but not limited to, one, three, five, or six. The first port 130 has the first electrical connector 112 at least partially disposed in the first port 130, and the second port 132 has the second electrical connector 114 at least partially disposed in the second port 132. The oil filter housing first portion first end 120 can also have a hose attachment point 134 in which extends away from the oil filter housing first portion first end 120. The hose attachment point 134 can be configured to receive a hose, not illustrated, to transfer oil from the oil filter assembly 100 to a motor or from the motor to the oil filter assembly 100. The oil filter housing first portion 102 can further have a channel 136 extending along an axis 138 within the oil filter housing intermediate portion 124 that oil can flow through. The oil filter housing first portion 102 can have any number of channels 136 extending along the axis 138 within the oil filter housing first portion intermediate portion 124 including, but not limited to, two, three, or four. The oil filter housing first portion 102 can be made from any desired material including, but not limited to metals, polymers, and ceramics.

[0036] The oil filter housing first portion intermediate portion 124 can have an oil filter housing first portion vertical channel 140 extending from the oil filter housing first portion first surface 126 and the oil filter housing first portion second surface 128. In the illustrated embodiment, the oil filter assembly 100 has four oil filter housing first portion vertical channels 140. However, the oil filter assembly 100 can have any number of oil filter housing first portion vertical channels 140 including, but not limited to, one, two, three, seven or ten. The oil filter housing first portion 102 can further have a plurality of retaining apertures 142 around an exterior perimeter 144 of the oil filter housing first portion 102. Each of the retaining apertures 142 is configured to receive a bolt. The retaining apertures 142 are configured to secure the oil filter assembly 100 in place. For example, the retaining apertures 142 can be used to secure the oil filter assembly 100 to a motor.

[0037] The oil filter housing first portion 102 further comprises the oil filter tube 110. The oil filter tube 110 can extend in a direction away from the oil filter housing first portion second surface 128. The oil filter tube 110 has an oil filter tube distal end 146 and an oil filter tube proximal end 148. The oil filter tube proximal end 148 can have an oil filter tube opening 150 having internal threads that are adapted to receive external threads of the cap 108. Although, in the illustrated embodiment, the oil filter tube opening 150

has internal threads and the cap **108** has external threads, in alternative embodiments the threads can be reversed. The cap **108** can be made from any desired material including, but not limited to, polymers and metals. The cap **108** can further have a filter attached thereto, and the filter is configured to be disposed within the oil filter tube **110** through the oil filter tube proximal end **148**. The oil filter assembly **100** can further have an oil filter tube cavity **152** that extends from the oil filter tube **110** in a direction towards the second surface of the oil filter housing first portion **102**. In the illustrated embodiment the oil filter tube cavity **152** has an O-ring that is configured to prevent oil from leaking from the oil filter assembly **100**. Also, the oil filter tube **110** is adapted to receive an oil filter. The oil filter can be attached to the cap **108**; however, it doesn't have to be. The cap **108** can also be made from a different material than the oil filter housing first portion **102** and the oil filter housing second portion **104**.

[0038] The oil filter housing first portion **102** further includes the first flow port **116** and the second flow port **118**. Although the illustrated embodiment only has two flow ports, any desired number of flow ports may be present including, but not limited to, one, three, four, six, or seven. The first flow port **116** and the second flow port **118** can serve as any desired functionality. For example, the first flow port **118** can transfer hot oil from the engine to be filtered and cooled prior to the return of cooled oil exiting the oil filter assembly **100** through the second flow port **118**. Alternatively, the first flow port **116** and the second flow port **118** need not be used. The first flow port **116** and the second flow port **118** can be plugged by using a first bolt **154** and a second bolt **156** to prevent oil from leaving the oil filter assembly **100** through the first flow port **116** and the second flow port **118**.

[0039] Each of FIGS. **5**, **6**, **7**, and **8** illustrates the oil filter housing second portion **104**. In the illustrated embodiment the oil filter housing second portion **104** is oval shaped; however, any desired shape is acceptable. The oil filter housing second portion **104** has an oil filter housing second portion first surface **158**, an oil filter housing second portion second surface **160** opposing the oil filter housing second portion first surface **158**, a perimeter wall **162**. The oil filter housing second portion **104** defining a first hole **164** and a second hole **166** each of which extend through the oil filter housing second portion **104** from the oil filter housing second portion first surface **158** to the oil filter housing second portion second surface **160**. In the illustrated embodiment, the perimeter wall **162** extends fully around the oil filter housing second portion **104** and orthogonally away from the oil filter housing second portion first surface **158**. The perimeter wall **162** has a perimeter wall height **170** that is approximately 5 mm; however, the perimeter wall **162** can have any perimeter wall height **170** including, but not limited to, 1 mm, 2 mm, 4 mm, 6 mm, or 10 mm. The oil filter housing second portion **104** further includes the first hole **164** and the second hole **166** that extend from the oil filter housing second portion first surface **158** to the oil filter housing second portion second surface **160**. In the illustrated embodiment, the first hole **164** has a first diameter **172**, and the second hole **166** has a second diameter **174**. The first diameter **172** being smaller than the second diameter **174**. The first diameter **172** is approximately 14 mm and the second diameter **174** is approximately 16 mm. However, any desired diameter can be used for the first hole **164** and the second hole **166**. Although the illustrated embodiment

shows that the first hole **164** and the second hole **166** are different sizes, in alternative embodiments the first hole **164** and the second hole **166** can be the same size. The oil filter housing second portion **104** has an oil filter housing second portion first width **176** and an oil filter housing second portion second width **178**. The oil filter housing second portion first width **176** being greater than the oil filter housing second portion second width **178**. Although this embodiment of the oil filter housing second portion **104** is illustrated in this manner, the oil filter housing second portion first width **176** and the oil filter housing second portion second width **178** can be the same. For example, oil filter housing second portion first widths **176** can include, but are not limited to, approximately 10 mm to approximately 25 mm. Example oil filter housing second portion second widths **178** can include, but are not limited to, approximately 20 mm to approximately 30 mm. The oil filter housing second portion **104** also has an oil filter housing second portion length **180** that is approximately 46 mm. However, the oil filter housing second portion length **180** can be any desired length including, but not limited to, approximately 45 mm to approximately 50 mm. The oil filter housing second portion **104** is configured to be disposed on the oil filter housing first portion **102** such that the first flow port **116** extends through the first hole **164**, the second flow port **118** extends through the second hole **166**, and the perimeter wall **168** extends toward the oil filter housing first portion **102**. A first washer **182** and a second washer **184** can be disposed between the oil filter housing second portion **104** and the oil filter housing first portion **102** to help prevent oil from leaking from the first flow port **116** and the second flow port **118**. The oil filter housing second portion **104** provides several benefits to the oil filter assembly **100**. To start with, the oil filter housing second portion **104** provides support to lines that can be installed from the motor to the oil filter assembly **100**. Further, the oil filter housing second portion **104** provides a means to keep lines together while attaching them to the first flow port **116** and the second flow port **118**. The lines are not illustrated; however, these lines can be replaced with the first bolt **154** and second bolt **156** when the first flow port **116** and the second flow port **118** are not in use. The oil filter housing second portion **104** can be made from any desired material including, but not limited to, metals, polymers, and ceramics. The material of the oil filter housing second portion **104** can be the same material, or a different material, than that of oil filter housing first portion **102**.

[0040] The oil filter assembly **100** further comprises the oil cooler **106**, best seen in FIGS. **1**, **2**, **3** and **4**, that is removably attached to the oil filter housing first portion first surface **126**. The oil cooler **106** has an oil cooler length **186**, an oil cooler width **188**, an oil cooler first surface **190**, and an oil cooler second surface **192**. The oil cooler **106** is removably attached to the oil filter housing first portion first surface **126** by a plurality of bolts; however, any means of attachment can be used to attach the oil cooler **106** to the oil filter housing first portion **102**. An oil cooler height **194** extends from the oil cooler first surface **190** to the oil cooler second surface **192**. The oil cooler **106** has at least one oil cooler vertical channel **196** extending from the oil cooler second surface **192** in a direction of the oil cooler first surface **190**. Although not illustrated, the illustrated embodiment has four oil cooler vertical channels **196**. These oil cooler vertical channels **196** allow oil to flow through the oil

cooler **106** to help lower the temperature of the oil. The oil cooler vertical channels **196** are configured to mate with the oil filter housing first portion vertical channels **140** to allow oil to flow to and from the oil filter housing first portion **102**.

[0041] Each of FIGS. **9**, **10**, **11**, and **12** illustrates a second example oil filter assembly **200**. The oil filter assembly **200** includes an oil filter housing **201**, an oil filter (not illustrated) disposed in an oil filter tube **210** defined by the oil filter housing **101**, and an oil cooler **206**. The oil filter housing **201** has an oil filter housing first portion **202** and an oil filter housing second portion **204**. The oil filter assembly **200** further has a cap **208**, a first electrical connector **212**, a second electrical connector **214**, a first flow port **216**, a second flow port **218**, and a third flow port **217**.

[0042] As illustrated, the oil filter housing first portion **202** has an oil filter housing first portion first end **220**, an oil filter housing first portion second end **222**, and an oil filter housing first portion intermediate portion **224** extending between the oil filter housing first portion first end **220** and the oil filter housing first portion second end **222**. The oil filter housing first portion **202** further has an oil filter housing first portion first surface **226** and an oil filter housing first portion second surface **228**. The oil filter housing first portion first end **220** comprises a first port **230** and a second port **232**. The oil filter housing first portion first end **220** can have as many ports as desired including, but not limited to, one, three, five, or six. The first port **230** has the first electrical connector **212** at least partially disposed in the first port **230**, and the second port **232** has the second electrical connector **214** at least partially disposed in the second port **232**. The oil filter housing first portion first end **220** can also have a hose attachment point **234** in which extends away from the oil filter housing first portion first end **220**. The hose attachment point **234** can be configured to receive a hose, not illustrated, to transfer oil from the oil filter assembly **200** to a motor or from a motor to the oil filter assembly **200**. The oil filter housing first portion **202** can further have a channel **236** extending along an axis **238** within the oil filter housing first portion intermediate portion **224** that oil can flow through. The oil filter housing first portion **202** can have any number of channels **236** extending along the axis **238** within the oil filter housing first portion intermediate portion **224** including, but not limited to, two, three, four, or five. The oil filter housing first portion **202** can be made from any desired material including, but not limited to metals, polymers, and ceramics.

[0043] The oil filter housing first portion intermediate portion **224** can have an oil filter housing first portion vertical channel **240** extending from the oil filter housing first portion first surface **226** and the oil filter housing first portion second surface **228**. In the illustrated embodiment, the oil filter assembly **200** has four oil filter housing first portion vertical channels **240**. However, The oil filter assembly **200** can have any number of oil filter housing first portion vertical channels **240** including, but not limited to, one, two, three, seven or ten. The oil filter housing first portion **202** can further have a plurality of retaining apertures **242** around an exterior perimeter **244**. Each of the retaining apertures **242** is configured to receive a bolt. The retaining apertures **242** are configured to secure the oil filter assembly **200** in place. For example, the retaining apertures **242** can be used to secure the oil filter assembly **200** to a motor.

[0044] The oil filter housing first portion **202** further comprises the oil filter tube **210**. The oil filter tube **210** can extend in a direction away from the oil filter housing first portion second surface **228**. The oil filter tube **210** has an oil filter tube distal end **246** and an oil filter tube proximal end **248**. The oil filter tube proximal end **248** can have an oil filter tube opening **250** having internal threads that are adapted to receive external threads of the cap **208**. Although in the illustrated embodiment the oil filter tube opening **250** has internal threads and the cap **208** has external threads, in alternative embodiments the threads can be reversed. The cap **208** can be made from any desired material including, but not limited to, polymers and metals. The cap **208** can further have a filter attached thereto, and the filter is configured to be disposed within the oil filter tube **210** through the oil filter tube proximal end **248**. The oil filter assembly **200** can further have an oil filter tube cavity **252** that extends from the oil filter tube **210** in a direction towards the second surface of the oil filter housing first portion **202**. In the illustrated embodiment, the oil filter tube cavity **252** has an O-ring that is configured to prevent oil from leaking from the oil filter assembly **200**. Also, the oil filter tube **210** is adapted to receive an oil filter. The oil filter can be attached to the cap **208**; however, it doesn't have to be. The cap **208** can also be made from a different material than the oil filter housing first portion **202** and the oil filter housing second portion **204**.

[0045] The oil filter housing first portion **202** further includes the first flow port **216**, the second flow port **218**, and the third flow port **217**. Although the illustrated embodiment has three flow ports, any desired number of flow ports may be present including, but not limited to, one, two, four, six, or seven. The first flow port **216**, the second flow port **218**, and third flow port **217** can serve as any desired functionality. For example, the first flow port can transfer hot oil from the engine to be filtered and cooled prior to the return of cooled oil exiting the oil filter assembly **200** through the second flow port **218** and third flow port **217**. Alternatively, the first flow port **216**, the second flow port **218**, or third flow port **217** need not be used. The first flow port **216**, the second flow port **218**, and third flow port **217** can be plugged by using a first bolt **254**, a second bolt **256**, and third bolt **255** to prevent oil from leaving the oil filter assembly **200**.

[0046] Each of FIGS. **13**, **14**, **15**, and **16** illustrates the oil filter housing second portion **204**. The oil filter housing second portion **204** can be any shape; however, in the illustrated embodiment the oil filter housing second portion **204** has a shape having three lobes. The oil filter housing second portion **204** has an oil filter housing second portion first surface, an oil filter housing second portion second surface **260** opposing the oil filter housing second portion first surface **258**, a perimeter wall **262**. The oil filter housing second portion **204** defining a first hole **264**, a second hole **266**, and a third hole **265** each of which extends through the oil housing second portion from the oil filter housing second portion first surface to the oil filter housing second portion second surface. In the illustrated embodiment, the perimeter wall **262** extends fully around the oil filter housing second portion **204** and orthogonally away from the oil filter housing second portion first surface **258**. The perimeter wall **262** has a perimeter wall height **270** that is approximately 5 mm; however, the perimeter wall can have any perimeter wall height **270** including, but not limited to, 1 mm, 2 mm, 4 mm, 6 mm, or 10 mm. The first hole **264**, a second hole **266**, and

the third hole **265** extend from the oil filter housing second portion first surface **258** to the oil filter housing second portion second surface **260**. In the illustrated embodiment, the first hole **264** has a first diameter **272**, the second hole **266** has a second diameter **274**, and the third hole **265** has a third diameter **273**. The first diameter **272**, the second diameter **274**, and the third diameter **273** are all different sizes to allow an individual to know what line, not illustrated, is to go in which hole. The first diameter **272** is approximately 14 mm, the second diameter **274** is approximately 16 mm, and the third diameter **273** is approximately 18 mm. Although the illustrated embodiment shows that the first hole **264**, the second hole **266**, and the third hole **265** are the same size, in alternative embodiments the first hole **264**, the second hole **266**, and the third hole **265** can be different sizes. Alternatively, two of the holes could be the same and one be different. The oil filter housing second portion **204** is configured to be disposed on the oil filter housing first portion **202** such that the first flow port **216** extends through the first hole **264**, the second flow port **218** extends through the second hole **266**, the third flow port **217** extends through the third hole **265**, and the perimeter wall **268** extends toward the oil filter housing first portion **202**. A first washer **282**, a second washer **284**, and a third washer **283** can be disposed between the oil filter housing second portion **204** and the oil filter housing first portion **202** to help prevent oil from leaking from the first flow port **216**, the second flow port **218**, and the third flow port **217**. The oil filter housing second portion **204** provides several benefits to the oil filter assembly **200**. To start with, the oil filter housing second portion **204** provides support to lines that can be installed from the motor to the oil filter assembly **200**. Further, the oil filter housing second portion **204** provides a means to keep lines together while attaching them to the first flow port **216**, the second flow port **218**, and the third flow port **217**. The lines are not illustrated; however, these lines can be replaced with the illustrated first bolt **254**, second bolt **256**, and third bolt **255** when the first flow port **216**, the second flow port **218**, and third flow port **217** are not in use. The oil filter housing second portion **204** can be made from any desired material including, but not limited to, metals, polymers, and ceramics. The material of the oil filter housing second portion **204** can be the same material, or a different material, than that of oil filter housing first portion **202**.

[0047] The oil filter assembly **200** further comprises the oil cooler **206**, best seen in FIGS. 9, 10, 11, and 12, that is removably attached to the first surface of the oil filter housing first portion **202**. The oil cooler **206** has an oil cooler length **286**, an oil cooler width **288**, an oil cooler first surface **290** and an oil cooler second surface **292**. The oil cooler **206** is removably attached to the oil filter housing first portion first surface **226** by a plurality of bolts; however, any means of attachment can be used to attach the oil cooler **206** to the oil filter housing first portion **202**. The oil cooler height **294** extending from the oil cooler first surface **290** to the oil cooler second surface **292**. The oil cooler **206** has at least one oil cooler vertical channel **296** extending from the oil cooler second surface **290** in a direction of the oil cooler first surface **292**. Although not illustrated, the illustrated embodiment has four oil cooler vertical channels **296**. These oil cooler vertical channels **296** allow oil to flow through the oil cooler **206** to help lower the temperature of the oil. The oil cooler vertical channels **296** are configured to mate with the

oil filter housing first portion vertical channels **240** to allow oil to flow to and from the oil filter housing first portion.

[0048] Each of FIGS. 17, 18, 19, and 20 illustrates a third example oil filter assembly **300**. The oil filter assembly **300** includes an oil filter housing **301**, an oil filter (not illustrated) disposed in an oil filter tube **310** defined by the oil filter housing **101**, and an oil cooler **306**. The oil filter housing **301** has an oil filter housing first portion **302** and an oil filter housing second portion **304**. The oil filter assembly **300** further has a cap **308**, a first electrical connector **312**, a second electrical connector **314**, and a singular flow port **316**.

[0049] As illustrated, the oil filter housing first portion **302** has an oil filter housing first portion first end **320**, an oil filter housing first portion second end **322**, and an oil filter housing first portion intermediate portion **324** extending between the oil filter housing first portion first end **320** and the oil filter housing first portion second end **322**. The oil filter housing first portion **302** further has an oil filter housing first portion first surface **326** and an oil filter housing first portion second surface **328**. The oil filter housing first portion first end **320** comprises a first port **330** and a second port **332**. The oil filter housing first portion first end **320** can have as many ports as desired including, but not limited to, one, three, five, or six. The first port **330** has the first electrical connector **312** at least partially disposed in the first port **330**, and the second port **332** has the second electrical connector **314** at least partially disposed in the second port **332**. The oil filter housing first portion first end **320** can also have a hose attachment point **334** in which extends away from the oil filter housing first portion first end **320**. The hose attachment point **334** can be configured to receive a hose, not illustrated, to transfer oil from the oil filter assembly **300** to a motor or from the motor to the oil filter assembly **300**. The oil filter housing first portion **302** can further have a channel **336** extending along an axis **338** within the oil filter housing first portion intermediate portion **324** that oil can flow through. The oil filter housing first portion **302** can have any number of channels **336** extending along the axis **338** within the oil filter housing first portion intermediate portion **324**. The oil filter housing first portion **302** can be made from any desired material including, but not limited to metals, polymers, and ceramics.

[0050] The oil filter housing first portion intermediate portion **324** can have an oil filter housing first portion vertical channel **340** extending from the oil filter housing first portion first surface **326** and the oil filter housing first portion second surface **328**. In the illustrated embodiment, the oil filter assembly **300** has four oil filter housing first portion vertical channels **340**. However, the oil filter assembly **300** can have any number of oil filter housing first portion vertical channels **340** including, but not limited to, one, two, three, seven or ten. The oil filter housing first portion **302** can further have a plurality of retaining apertures **342** around an exterior perimeter **344**. Each of the retaining apertures **342** is configured to receive a bolt. The retaining apertures **342** are configured to secure the oil filter assembly **300** in place. For example, the retaining apertures **342** can be used to secure the oil filter assembly **300** to a motor.

[0051] The oil filter housing first portion **302** further comprises the oil filter tube **310**. The oil filter tube **310** can extend in a direction away from the oil filter housing first portion second surface **328**. The oil filter tube **310** has an oil

filter tube distal end 346 and an oil filter tube proximal end 348. The oil filter tube proximal end 348 can have an oil filter tube opening 350 having internal threads that are adapted to receive external threads of the cap 308. Although in the illustrated embodiment the oil filter tube opening 350 has internal threads and the cap 308 has external threads, in alternative embodiments the threads can be reversed. The cap 308 can be made from any desired material including, but not limited to, polymers and metals. The cap 308 can further have a filter attached thereto, and the filter is configured to be disposed within the oil filter tube 310 through the oil filter tube proximal end 348. The oil filter assembly 300 can further have an oil filter tube cavity 352 that extends from the oil filter tube 310 in a direction towards the second surface of the oil filter housing first portion 302. In the illustrated embodiment the oil filter tube cavity 352 has an O-ring that is configured to prevent oil from leaking from the oil filter assembly 300. Also, the oil filter tube 310 is adapted to receive an oil filter. The oil filter can be attached to the cap 308; however, it doesn't have to be. The cap 308 can also be made from a different material than the oil filter housing first portion 302 and the oil filter housing second portion 304.

[0052] The oil filter housing first portion 302 further includes the flow port 316. Although the illustrated embodiment only has one flow port 316, any desired number flow ports may be present including, but not limited to, two, three, four, six, or seven. The flow port 316 can serve as any desired functionality. For example, the flow port can transfer hot oil from the engine to be filtered and cooled, or the flow port can transfer cooled oil from the oil filter assembly 300 to a motor. Alternatively, the flow port 316 need not be used. The flow port 316 can be plugged by using a bolt 354 to prevent oil from leaving the oil filter assembly 300 through the flow port 316.

[0053] Each of FIGS. 21, 22, 23, and 24 illustrates the oil filter housing second portion 304. The oil filter housing second portion 304 is circular shaped; however, any desired shape can be used. The oil filter housing second portion 304 has an oil filter housing second portion first surface 358, an oil filter housing second portion second surface 360 opposing the oil filter housing second portion first surface 358, a perimeter wall 362. The oil filter housing second portion 304 defining a hole 364 which extends through the oil filter housing second portion 304 from the oil filter housing second portion first surface 358 to the oil filter housing second portion second surface 360. In the illustrated embodiment, the perimeter wall 362 extends fully around the oil filter housing second portion 304 and orthogonally away from the oil filter housing second portion first surface 358. The perimeter wall 362 has a perimeter wall height 370 that is approximately 5 mm; however, the perimeter wall 370 can have any perimeter wall height 370 including, but not limited to, 1 mm, 2 mm, 4 mm, 6 mm, or 10 mm. The oil filter housing second portion 304 further has a hole 364 that extends from the oil filter housing second portion first surface 358 to the oil filter housing second portion second surface 360. In the illustrated embodiment, the hole 364 has a diameter 372. The diameter 372 in the illustrated embodiment is approximately 14 mm. However, any desired size can be used including, but not limited to, 9 mm, 10 mm, 12 mm, 15 mm, and 20 mm. The oil filter housing second portion 304 has a diameter 372 that is approximately 22 mm; however, the oil filter housing second portion 304 can have any desired diameter 372. The oil filter housing second

portion 304 is configured to be disposed on the oil filter housing first portion 302 such that the flow port 316 extends through the hole 364 and the perimeter wall 368 extends toward the oil filter housing first portion 302. A washer 382 can be disposed between the oil filter housing second portion 304 and the oil filter housing first portion 302 to help prevent oil from leaking from the flow port 316. The oil filter housing second portion 304 provides several benefits to the oil filter assembly 300. To start with, the oil filter housing second portion 304 provides support to a line that can be installed from the motor to the oil filter assembly 300. The line is not illustrated; however, this line can be replaced with the illustrated bolt 354 when the flow port 316 is not in use. The oil filter housing second portion 304 can be made from any desired material including, but not limited to, metals, polymers, and ceramics. The material of the oil filter housing second portion 304 can be the same material, or a different material, than that of oil filter housing first portion 302.

[0054] The oil filter assembly 300 further comprises the oil cooler 306, best seen in FIGS. 17, 18, 19, and 20, that is removably attached to the oil filter housing first portion first surface 326. The oil cooler 306 has an oil cooler length 386, an oil cooler width 388, an oil cooler first surface 390 and an oil cooler second surface 392. The oil filter housing second portion 304 is removably attached to the first surface of the oil filter housing first portion by a bolt; however, any means of attachment can be used to attach the oil cooler 306 to the oil filter housing second portion 304. The oil cooler height 394 extending from the oil cooler first surface 390 to the oil cooler second surface 392. The oil cooler 306 has at least one oil cooler vertical channel 396 extending from the oil cooler second surface 392 in a direction of the oil cooler first surface 390. Although not illustrated, the illustrated embodiment has four oil cooler vertical channels 396. These channels allow oil to flow through the oil cooler 306 to help lower the temperature of the oil. The oil cooler vertical channels 396 are configured to mate with the oil filter housing first portion vertical channels 340 to allow oil to flow to and from the oil filter housing first portion 302.

[0055] Those with ordinary skill in the art will appreciate that various modifications and alternatives for the described and illustrated examples can be developed in light of the overall teachings of the disclosure, and that the various elements and features of one example described and illustrated herein can be combined with various elements and features of another example without departing from the scope of the invention. Accordingly, the particular examples disclosed herein have been selected by the inventors simply to describe and illustrate examples of the invention and are not intended to limit the scope of the invention or its protection, which is to be given the full breadth of the appended claims and any and all equivalents thereof.

We claim:

1. An oil filter assembly, comprising:

an oil filter housing first portion having an oil filter housing first portion first end, an opposing oil filter housing first portion second end, an oil filter housing first portion first surface and an opposing oil filter housing first portion second surface, the oil filter housing first portion first end defining an oil filter tube, the oil filter housing first portion second end defining first and second flow ports, the oil filter housing first portion first surface adapted for a secured connection with an internal combustion engine;

- an oil filter housing second portion having an oil filter housing second portion first surface, an oil filter housing second portion second surface opposing the oil filter housing second portion first surface, the oil filter housing second portion defining first and second holes extending through the oil filter housing second portion from the oil filter housing second portion first surface to the oil filter housing second portion second surface, the oil filter housing second portion disposed on the oil filter housing first portion such that the first flow port extends through the first hole, and the second flow port extends through the second hole;
- an oil filter adapted to be disposed within the oil filter tube; and
- an oil cooler adapted to be secured to the oil filter housing first portion first surface.
2. The oil filter assembly of claim 1, further comprising a third flow port on the oil filter housing first portion second end.
 3. The oil filter assembly of claim 2, further comprising a third hole extending from the oil filter housing second portion first surface through the oil filter housing second portion second surface.
 4. The oil filter assembly of claim 3, wherein the oil filter housing second portion is disposed over the first flow port, the second flow port, and the third flow port.
 5. The oil filter assembly of claim 1, further comprising a cap that is configured to attach to the oil filter tube.
 6. The oil filter assembly of claim 5, wherein the oil filter is attached to the cap.
 7. The oil filter assembly of claim 5, wherein the cap is made from a first material and the oil filter housing first portion is made from a second material, the first material being different from the second material.
 8. The oil filter assembly of claim 1, wherein the oil filter tube extends in a direction away from the oil filter housing first portion second surface.
 9. The oil filter assembly of claim 1, wherein the oil filter tube has an oil filter tube proximal end having an oil filter tube opening that is adapted to receive a cap.
 10. The oil filter assembly of claim 1, wherein the first hole has a first diameter and the second hole has a second diameter; and
 - wherein the first diameter and the second diameter are different.
 11. The oil filter assembly of claim 1, wherein the first hole has a first diameter and the second hole has a second diameter; and
 - wherein the first diameter and the second diameter are approximately the same.
 12. The oil filter assembly of claim 3, wherein the first hole has a first diameter, the second hole has a second diameter, and the third hole has a third diameter; and
 - wherein the first diameter, the second diameter, and the third diameter are different.
 13. The oil filter assembly of claim 3, wherein the first hole has a first diameter, the second hole has a second diameter, and the third hole has a third diameter; and
 - wherein the first diameter, the second diameter, and the third diameter are approximately the same.
 14. The oil filter assembly of claim 3, wherein the first hole has a first diameter, the second hole has a second diameter, and the third hole has a third diameter; and

wherein at least one of the first diameter, second diameter, or third diameters are different.

15. The oil filter assembly of claim 1, wherein the oil filter housing second portion has an oil filter housing second portion first width and an oil filter housing second portion second width.

16. The oil filter assembly of claim 1, wherein the oil cooler is removably attached to the oil filter housing first portion.

17. The oil filter assembly of claim 1, wherein the oil cooler has an oil cooler first surface and an oil cooler second surface; and

wherein the oil cooler has at least one channel extending from the oil cooler second surface in a direction of the oil cooler first surface.

18. The oil filter assembly of claim 1, wherein the oil cooler has an oil cooler first surface and an oil cooler second surface; and

wherein the oil cooler has a plurality of oil cooler vertical channels extending from the oil cooler second surface in a direction of the oil cooler first surface.

19. An oil filter assembly, comprising:

an oil filter housing first portion having an oil filter housing first portion first end, an opposing oil filter housing first portion second end, an oil filter housing first portion first surface, and an opposing oil filter housing first portion second surface, the oil filter housing first portion first end defining an oil filter tube, the oil filter housing first portion second end defining a flow port, the oil filter housing first portion first surface adapted for a secured connection with an internal combustion engine;

an oil filter housing second portion having an oil filter housing second portion first surface, an opposing oil filter housing second portion second surface opposing the oil filter housing second portion first surface, the oil filter housing second portion having an oil filter housing second portion first width and an oil filter housing second portion second width, the oil filter housing second portion defining a hole extending through the oil filter housing second portion from the oil filter housing second portion first surface to the oil filter housing second portion second surface, and the oil filter housing second portion disposed on the oil filter housing first portion such that the flow port extends through the hole;

an oil cooler having an oil cooler first surface and an oil cooler second surface removably attached to the oil filter housing first portion first surface; and

an oil filter adapted to be disposed within the oil filter tube.

20. An oil filter assembly, comprising:

an oil filter housing first portion having an oil filter housing first portion first end and an opposing oil filter housing first portion second end, an oil filter housing first portion first surface and an opposing oil filter housing first portion second surface, the oil filter housing first portion first end defining an oil filter tube, the oil filter housing first portion second end defining a flow port, the first surface adapted for secured connection with an internal combustion engine, the oil filter housing first portion having an oil filter housing first portion vertical channel extending from the oil filter

housing first portion first surface to the oil filter housing first portion second surface;

an oil filter housing second portion having an oil filter housing second portion first surface, an oil filter housing second portion second surface opposing the oil filter housing second portion first surface, the oil filter housing second portion having an oil filter housing second portion first width and an oil filter housing second portion second width, the oil filter housing second portion first width being different from the oil filter housing second portion second width, the oil filter housing second portion defining a hole extending through the oil filter housing second portion from the oil filter housing second portion first surface to the oil filter housing second portion second surface, and the oil filter housing second portion disposed on the oil filter housing first portion such that the flow port extends through the hole;

an oil cooler having an oil cooler first surface and an oil cooler second surface removably attached to the oil filter housing first portion first surface, the oil cooler having an oil cooler vertical channel extending from the oil cooler second surface in a direction toward the oil cooler first surface; and

an oil filter adapted to be disposed within the oil filter tube.

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